

Data Dictionary

Study: Determinants of Hypertension Treatment Uptake among Adults attending Primary Healthcare Facilities

Design: Multicentre cross-sectional study, 6 primary healthcare facilities, 1,500 adult attendees.

Files: hypertension_phc_raw.csv / .xlsx (used Days 1-2), hypertension_phc_clean.csv (tidy reference).

Primary outcome: treatment_uptake (currently on antihypertensive therapy), analysed among patients with htn_diagnosed = Yes.

#	Variable	Label	Type	Units / Coding	Role	Notes
1	patient_id	Patient identifier	Character	PHC-0001 ... PHC-1500	ID	Unique per patient. Raw file contains 3 duplicate records to be removed.
2	facility	Healthcare facility	Categorical	6 PHC facilities	Cluster/covariate	Multicentre design. Some values have leading/trailing spaces in raw file.
3	enroll_date	Date of enrolment	Date	Mixed: YYYY-MM-DD, DD/MM/YYYY, D-Mon-YYYY	Metadata	Mixed formats in raw file; parse on Day 2.
4	age	Age	Numeric	Years	Predictor	Implausible values (0, 200) present in raw file.
5	sex	Sex	Categorical	Female / Male	Predictor	Raw file mixes F/f/female and M/m/male spellings.
6	residence	Place of residence	Categorical	Urban / Rural	Predictor	Determinant of uptake.
7	education	Highest education	Ordinal	None < Primary < Secondary < Tertiary	Predictor	Set as ordered factor. Some blanks in raw file.
8	occupation	Occupation	Categorical	Unemployed/ Farmer/Trade r/Professional /Other	Predictor	
9	marital_status	Marital status	Categorical	Single/Married/Divorced/Widowed	Predictor	

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10	health_insurance	Has health insurance	Binary	Yes / No	Predictor	Raw file mixes Yes/No /Y/N/1/0. Determinant of uptake.
11	height_cm	Height	Numeric	centimetres	Derived input	Used to compute BMI. One decimal-point error (17).
12	weight_kg	Weight	Numeric	kilograms	Derived input	Missing sentinels (NA, blank). One impossible value (7).
13	bmi	Body mass index	Numeric	kg/m ²	Predictor	Provided but contains errors/missing; recompute from height & weight.
14	smoking	Smoking status	Categorical	Never / Former / Current	Predictor	Some missing in raw file.
15	alcohol	Alcohol intake	Categorical	None / Moderate / Heavy	Predictor	
16	physical_activity	Physical activity level	Ordinal	Low / Moderate / High	Predictor	
17	family_history_htn	Family history of hypertension	Binary	Yes / No	Predictor	Mixed coding in raw file. Determinant of uptake.
18	diabetes	Diabetes mellitus	Binary	Yes / No	Predictor	Mixed coding in raw file. Strong determinant of uptake.
19	sbp_mmhg	Systolic blood pressure	Numeric	mmHg	Clinical	Implausible values (0, 700) present in raw file.
20	dbp_mmhg	Diastolic blood pressure	Numeric	mmHg	Clinical	Implausible value (5) present in raw file.
21	total_cholesterol_mmol_l	Total cholesterol	Numeric	mmol/L	Lab biomarker	Missing sentinels: NA, blank, -99.
22	hdl_mmol_l	HDL cholesterol	Numeric	mmol/L	Lab biomarker	

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23	ldl_mmol_l	LDL cholesterol	Numeric	mmol/L	Lab biomarker	Missing sentinels present.
24	triglycerides_mmol_l	Triglycerides	Numeric	mmol/L	Lab biomarker	
25	fasting_glucose_mmol_l	Fasting glucose	Numeric	mmol/L	Lab biomarker	Missing sentinel 999 present.
26	creatinine_umol_l	Serum creatinine	Numeric	umol/L	Lab biomarker	
27	sodium_mmol_l	Serum sodium	Numeric	mmol/L	Lab biomarker	
28	potassium_mmol_l	Serum potassium	Numeric	mmol/L	Lab biomarker	
29	knowledge_score	Hypertension knowledge score	Numeric	0-20	Predictor	Higher = better knowledge. Determinant of uptake.
30	distance_to_facility_km	Distance to facility	Numeric	kilometres	Predictor	Access barrier. Some missing.
31	comorbidity_count	Number of comorbidities	Count	0+	Predictor	
32	htn_diagnosed	Diagnosed hypertensive	Binary	Yes / No	Filter	Defines analysis population for uptake (Days 4-5).
33	months_since_diagnosis	Months since HTN diagnosis	Numeric	Months	Predictor	0 if not diagnosed.
34	treatment_uptake	On antihypertensive treatment	Binary	Yes / No	PRIMARY OUTCOME	Mixed coding in raw file. Analysed among diagnosed patients.
35	adherence	Treatment adherence	Categorical	Good / Poor (blank if untreated)	Secondary outcome	Only defined where treatment_uptake = Yes.
36	bp_controlled	Blood pressure controlled	Binary	Yes / No (blank if untreated)	Secondary outcome	SBP<140 and DBP<90 among treated.

Known data-quality issues in the raw file (for the Day 2 cleaning exercise)

- **Inconsistent category spellings:** sex coded as Female/F/female/f and Male/M/male/m.
- **Mixed binary codings:** diabetes, family_history_htn, health_insurance, htn_diagnosed, treatment_uptake mix Yes/No, Y/N and 1/0.
- **Missing-value sentinels:** blank, NA, 999, -99 in several numeric columns.
- **Implausible values:** age 0 and 200; sbp_mmhg 0 and 700; weight_kg 7; height_cm 17; dbp_mmhg 5.
- **Whitespace:** leading/trailing spaces in some facility, residence, education, occupation values.
- **Duplicates:** 3 duplicate patient records (1,503 rows, 1,500 unique IDs).
- **Mixed date formats** in enroll_date.